

SAMUTPRAKAN AGROMET, Thailand

WMO: 484200

Lat: 13.517N

Lon: 100.762E

Elev: 10

StdP: 14.69

Time Zone: 7.00 (E07)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	67.3	70.2	55.0	64.4	72.0	58.5	73.1	75.3	18.4	77.3	16.4	80.3	4.1	40	0.409

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	7.6	92.6	79.5	91.7	79.7	91.0	79.7	83.1	89.6	82.5	89.0	81.9	88.4	6.8	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
81.2	162.2	88.3	80.6	159.2	87.6	80.1	156.0	87.0	47.1	89.9	46.4	89.2	45.8	88.6	85.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
17.4	15.5	13.8	DB	64.3	95.7	3.3	2.3	61.9	97.3	60.0	98.7	58.1	100.0	55.8	101.6	(4)
			WB	57.8	84.3	3.9	1.0	55.0	85.0	52.7	85.6	50.6	86.2	47.8	87.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	83.8	80.2	82.6	84.5	86.2	86.4	85.2	84.4	84.3	83.8	83.9	83.2	80.7
	DBStd	2.98	3.42	2.24	2.39	2.16	2.17	1.53	1.70	1.58	1.72	1.96	2.43	3.71
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	12337	937	914	1069	1087	1128	1057	1067	1064	1015	1050	996	953
	CDD65	6862	472	494	604	637	663	607	602	599	565	585	546	488
	CDH74	84725	4685	5617	7582	8617	9155	8239	7778	7725	6999	7011	6297	5020
	CDH80	36395	1508	1986	3308	4375	4723	3954	3373	3339	2821	2768	2442	1798
Wind	WSAvg	6.2	4.5	6.5	7.8	7.8	7.0	7.9	7.3	7.0	5.3	3.8	4.2	4.5
Precipitation	PrecAvg	58.5	0.7	0.7	1.9	3.3	7.6	7.0	6.6	7.8	12.0	9.1	1.5	0.3
	PrecMax	76.5	3.4	2.6	6.1	8.6	15.6	12.8	10.5	13.7	18.5	16.5	5.6	1.7
	PrecMin	40.0	0.0	0.0	0.2	0.6	2.4	2.4	1.7	3.7	8.0	2.2	0.0	0.0
	PrecStd	8.7	0.8	0.7	1.7	2.4	3.3	2.6	2.4	3.0	2.5	3.1	1.5	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	90.9	91.8	92.7	93.4	93.5	91.7	90.8	91.6	92.4	92.5	92.6	93.0
		MCWB	75.5	74.1	76.6	80.1	81.3	80.9	79.3	80.1	79.9	79.4	78.3	77.7
	2%	DB	89.1	89.5	90.3	91.9	92.1	90.6	89.6	89.5	90.3	91.1	91.3	91.0
		MCWB	74.8	74.9	78.6	80.8	81.3	80.7	79.7	79.8	80.2	79.4	77.9	77.0
	5%	DB	87.8	87.9	89.0	90.9	91.3	89.7	88.8	88.6	89.0	89.9	90.1	89.2
		MCWB	75.5	77.1	79.5	81.0	81.3	80.4	79.5	79.5	79.8	79.3	78.0	76.6
	10%	DB	86.5	86.9	88.1	90.0	90.4	88.9	87.9	87.8	88.1	88.7	88.9	87.7
		MCWB	75.8	77.6	79.7	81.1	81.0	80.2	79.3	79.3	79.5	79.2	77.8	76.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	80.6	80.9	82.4	84.0	84.3	82.7	81.6	81.8	82.4	81.8	81.8	81.0
		MCDB	86.2	86.7	88.8	90.5	90.8	89.1	86.8	87.6	89.4	88.8	88.3	87.5
	2%	WB	79.7	80.1	81.7	83.2	83.2	81.9	80.9	81.0	81.3	81.0	81.0	80.0
		MCDB	85.2	85.8	87.9	89.7	89.8	88.2	86.8	87.0	87.8	87.7	87.4	86.1
	5%	WB	79.0	79.7	81.0	82.5	82.6	81.4	80.5	80.6	80.7	80.6	80.3	79.1
		MCDB	84.4	85.4	87.1	89.1	89.0	87.7	86.8	86.8	86.9	87.1	86.7	85.2
	10%	WB	78.3	79.2	80.6	81.9	81.9	80.9	80.0	80.1	80.1	80.1	79.7	78.3
		MCDB	83.5	84.7	86.6	88.4	88.5	87.2	86.3	86.3	86.2	86.3	85.8	84.6
Mean Daily Temperature Range	5% DB	MDBR	12.3	8.9	7.3	7.4	7.6	6.9	6.6	7.0	8.2	9.4	10.6	12.6
		MCDBR	13.4	11.1	8.7	8.0	8.0	8.3	7.3	7.8	9.5	10.7	11.9	13.6
		MCWBR	4.7	4.3	4.0	3.7	3.5	3.6	3.4	3.6	4.0	3.6	3.8	4.5
	5% WB	MCDBR	8.9	6.9	6.6	6.5	7.3	8.0	7.0	7.6	8.9	9.9	10.3	10.8
	MCWBR	3.6	3.2	3.6	3.5	3.7	3.8	3.6	3.7	4.0	3.9	3.8	4.1	
Clear-Sky Solar Irradiance	taub	0.542	0.608	0.630	0.570	0.496	0.467	0.461	0.463	0.484	0.570	0.518	0.503	
	taud	1.970	1.835	1.819	2.003	2.264	2.345	2.361	2.343	2.272	1.987	2.114	2.108	
	Ebn at Noon	241	231	230	243	257	262	264	267	263	237	246	248	
	Edh at Noon	57	67	70	58	44	40	39	41	44	57	49	49	
All-Sky Solar Radiation	RadAvg	1619	1744	1780	1898	1719	1643	1575	1596	1546	1511	1561	1561	
	RadStd	95	77	128	120	116	82	81	97	86	83	107	76	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	Regional (31 neighbors)	N/A	N/A	N/A	N/A	+0.36	+0.38	N/A	N/A	+126	+126

Nomenclature: See separate page